

Akka vs. AWS Bedrock (AgentCore)

A comparison for teams building agentic AI

July 2026

AWS Bedrock AgentCore is a set of separate AWS services you wire and operate yourself — not an integrated, governed platform. AgentCore gives you seven modular building blocks (Runtime, Memory, Gateway, Identity, Code Interpreter, Browser, Observability) that you assemble on top of Bedrock model access, Guardrails, and Knowledge Bases — each billed and operated separately, all running inside AWS. Akka delivers orchestration, agents, memory, streaming, APIs, observability, and governance as one runtime, with a six-nines SLA Akka owns, deployable on any cloud — and Akka Specify can deliver and run the entire governed system for you as a guaranteed outcome.

99.9999%

AKKA AVAILABILITY SLA

Weeks

AKKA SPECIFY DELIVERY

Fixed

ONE ALL-IN PRICE

90%

LESS INFRASTRUCTURE

DIMENSION	AWS BEDROCK AGENTCORE	AKKA
What it is	A set of modular AWS services for building and hosting agents	A full-stack agentic systems platform
Scope	Seven AgentCore modules assembled on Bedrock model access, Guardrails, and Knowledge Bases — the customer wires and operates the seams	Orchestration, agents, memory, streaming, APIs, observability, and governance on one runtime
How you reach production	Self-integrate the seven AgentCore services, or contract a systems-integration engagement	Build with spec-driven development, or Akka Specify delivers and runs it
Commercial model	Consumption-metered across the AgentCore services and model inference, plus integration labor	One fixed price — platform, infrastructure, tokens, training, delivery, and operations
Outcome guarantee	Effort only; the outcome is not guaranteed	The delivered outcome is guaranteed
Availability SLA	99.9% monthly for the Bedrock APIs, measured by request errors; service credits only	99.9999% — entire platform, backed by indemnities
RTO / RPO	Not published for the agent layer; inherits the AWS regional posture the customer architects	Sub-1-minute RTO; zero-byte RPO; active-active across regions
Deployment	AWS only — IAM, VPC/PrivateLink, CloudFormation, CloudWatch	Any cloud (AWS/Azure/GCP VPC), own Kubernetes, on-prem, sovereign cloud
Memory	AgentCore Memory — a separate, separately-billed service	Durable in-memory, 4ms reads / sub-10ms writes, built in
Governance / EU AI Act	Guardrails (model layer) + AgentCore Policy (tool layer), assembled; no embedded regulation/control enforcement	Aspect-woven runtime enforcement + pre-production classification against 186 regs / 877 controls
Cost model	Consumption-metered across the seven AgentCore services (per-vCPU-hour, per-request, per-token) plus separately billed model inference	Shared compute; up to 90% lower infrastructure for the same workload, one fixed price
Maturity	GA October 2025; Evaluations GA March 2026; Policy + Guardrails GA June 2026 — capabilities still landing	18 years; 100,000+ production deployments; 52 banks

Improves after go-live	AgentCore Evaluations detects quality drift; retraining and distillation are not provided	Continuous evaluation, reinforcement learning, and distillation — up to 80% lower token cost over time
Certifications	Bedrock: ISO, SOC, CSA STAR L2, GDPR, FedRAMP High, HIPAA-eligible	19 standards (SOC 2 II + public SOC 3, ISO 27001/42001, EU AI Act, NIST AI RMF)

1. AgentCore Is a Set of Services to Assemble; Akka Is One Platform

AWS Bedrock AgentCore is a collection of separate AWS services the customer integrates and operates, not an integrated platform. At [general availability](#), it ships seven modular services — Runtime, Memory, Gateway, Identity, Code Interpreter, Browser, and Observability — and an agent of substance also draws on Bedrock model access, Bedrock Guardrails, and Knowledge Bases. Each is provisioned, secured, billed, and operated on its own; the customer owns every seam between them.

Capability	AWS Bedrock AgentCore	Akka
Agent hosting / runtime	AgentCore Runtime (separate service)	Built in
Durable memory	AgentCore Memory (separate, separately billed)	Built in, 4ms / sub-10ms
Tool / API gateway	AgentCore Gateway (separate service)	Built in
Identity & authorization	AgentCore Identity (separate service)	Built in
Observability	AgentCore Observability on CloudWatch (separate)	Built in
Model access	Amazon Bedrock (separately billed per token)	Bring any model
Real-time streaming	Not a native AgentCore service	Built in, backpressured, petabyte-scale
Governance / policy enforcement	Policy + Guardrails, assembled	Inline, runtime-embedded
Delivery & operation	The customer integrates and operates all seven services, or contracts a systems-integration engagement	Delivered and operated by Akka Specify as a guaranteed outcome, in weeks

Akka delivers all of this on one runtime with one operational model. There are no seams to integrate, no per-service SLAs to reconcile, and no cross-service failure modes the customer inherits — and, through Akka Specify, Akka will integrate, deliver, and run the whole system for you.

2. Two Ways to Production: Assemble It Yourself, or Have It Delivered

Reaching a production agentic system on AgentCore means integrating seven separate services — Runtime, Memory, Gateway, Identity, Code Interpreter, Browser, and Observability — plus Bedrock model access, Guardrails, and Knowledge Bases, and operating every seam between them. You do that with your own engineers, or you contract a systems-integration engagement to assemble it for you — either way, there is no single vendor accountable for the finished, governed system. Akka offers a second path entirely: **Akka Specify** takes your specifications and delivers and operates the whole governed system for you as a guaranteed outcome — in weeks, for one fixed price.

	With AWS Bedrock AgentCore	With Akka Specify
Model	Self-integrate the seven AgentCore services, or a systems-integration engagement	You provide the specifications
Who does the work	Your engineers, or forward-deployed / staff-augmentation contractors	Akka generates, governs, delivers, and runs the system
Timeline	Months, by construction	Weeks, not quarters
Billing	Consumption metering across seven services plus model spend, plus integration labor	One fixed price
Afterward	You own and operate every service and every seam	Kept up, safe, and improving — operated by Akka
Guarantee	Effort is guaranteed; the outcome is not	The delivered outcome is guaranteed

Under AWS's own Shared Responsibility Model, the availability, integration, and operation of everything built on top of Bedrock's managed infrastructure are the customer's. A team that cannot self-integrate contracts a systems-integration engagement — billed for time and materials, guaranteeing effort rather than the outcome, and handing back a system the team then owns and operates. Akka

Specify inverts that: you provide a handful of plain-language specifications, and Akka generates, tests, governs, deploys, and runs the system — then keeps it available, safe, and improving under one agreement.

3. Availability and Disaster Recovery

The agent layer has no dedicated availability SLA. Amazon Bedrock publishes a [99.9% monthly uptime](#) commitment, measured as the percentage of Bedrock API requests that do not fail with errors, calculated per region, with service credits as the sole remedy. AgentCore inherits the regional, single-cloud posture the customer architects on top of AWS — the customer owns multi-region failover, RTO, and RPO for the assembled system.

Metric	AWS Bedrock AgentCore	Akka
Availability SLA	99.9% monthly (Bedrock API errors)	99.9999%
Allowed downtime / year	~8.8 hours	~31 seconds
RTO	Customer-architected	Sub-1 minute
RPO	Customer-architected	Zero byte
SLA scope	The Bedrock model APIs	The entire platform
Who operates it	The customer (the self-integrated services)	Akka SREs, 24/7
Remedy	Service credits	Contractual indemnities

99.9% per region versus 99.9999% is the difference between roughly 8.8 hours and 31 seconds of allowed downtime a year. Akka's SLA covers the whole running system — orchestration, agents, memory, streaming, APIs, and governance — and Akka owns *and operates* it with 24/7 SRE.

4. Cheaper to Operate — and It Keeps Getting Cheaper

AI systems built with Akka are up to **90% cheaper to operate** than Python-based systems — a function of the infrastructure required for the same agentic transaction volume, not a list-price comparison. AgentCore meters its services independently — Runtime, Memory, Gateway, Identity, Code Interpreter, Browser, and Observability, plus Evaluations and Policy — on per-vCPU-hour, per-request, and per-token units, on top of standard Bedrock model token spend. Every layer the agent touches is a separate consumption meter that moves with load.

Akka runs all of it on one shared-compute runtime. The efficiency comes from actor concurrency (~10T tokens/core/year vs. ~2T; ~80% less compute than Python frameworks; Manulife: up to 300% more concurrency, 30–50% faster processing after porting from Python), shared compute, and micro-checkpointing that minimizes retries. The spend is one fixed price — not a stack of usage meters that scale with traffic.

And the cost falls after go-live

The delivered outcome compounds. Akka Verify runs continuous evaluation, reinforcement learning on production and synthetic data, and distillation to smaller specialized models — cutting token cost up to 80% while raising accuracy over time. AgentCore Evaluations detects quality drift; it does not retrain, distill, or ship the improvement. With Akka Specify, that tuning is part of the operated outcome, not a project the customer runs later.

The EU AI Act Penalties Are Enforceable Now

What a violation costs

Violation	Maximum Fine
Prohibited AI practices (Art. 5)	EUR 35M or 7% global turnover
High-risk obligations (Art. 9-15)	EUR 15M or 3% global turnover
Incorrect information (Art. 99)	EUR 7.5M or 1.5% global turnover

High-risk AI carries a 10-year logging-retention obligation (Art. 72), enforceable since Feb 2025 (prohibited) / Aug 2025 (high-risk).

5. Governance and the EU AI Act

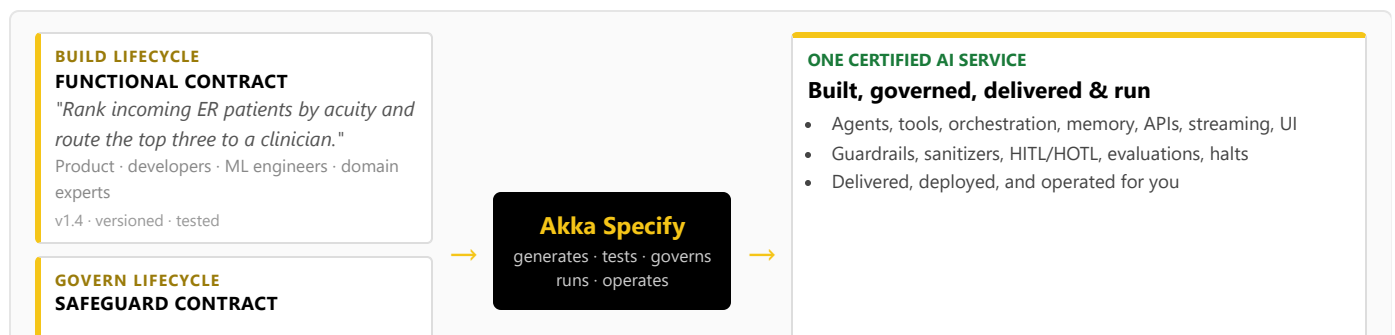
AgentCore offers governance as assembled services, not embedded enforcement of regulation. [Bedrock Guardrails](#) operate at the model-inference layer (unsafe content, PII); AgentCore Policy operates at the tool-access layer. Both reached [GA in policy in June 2026](#). They enforce vendor-defined safety filters and tool rules — not the obligations of a named regulation. There is no built-in classification against the EU AI Act, no immutable hash-chained evidence ledger as a platform guarantee, no pre-deployment governance gate, and no sealed audit artifact.

How Akka governs

At the runtime: inline guardrails, policies, LLMs-as-a-judge, and sanitizers; hash-chained immutable evidence; HITL/HOTL control; atomic PII scrub-with-explain; pre-deployment classification against **186 regulations and 877 controls — 288 of which carry a financial penalty**; a multi-persona sign-off recipe engine; a sealed Governance Posture Package; and Akka Verify proving conformance from the running system. The governance lifecycle is versioned and tested independently of the build.

6. One Certified System — Built, Governed, Delivered, and Run

Building on AgentCore means engineers wiring and operating a set of AWS services; there is no first-class path for a product manager, domain expert, or risk officer to contribute, and governance is assembled from separate services rather than running as its own lifecycle. Akka runs two independent lifecycles on one platform via **Akka Specify**:



"Block prohibited practices under EU AI Act Article 5; notify regulators within 24h of any incident."

Risk · security · compliance

v2.1 · versioned & tested independent of the build

Akka Verify  validates the running system against both specs and fine-tunes the AI from production data.

The build lifecycle and the governance lifecycle are versioned and tested independently, by different audiences — an audience and a workflow that a set of assembled cloud services has no equivalent for. Akka generates, tests, governs, **runs, and operates** one certified AI service from both specs, and through **Akka Specify**, that certified system is delivered and operated as a guaranteed outcome — a delivery model AgentCore has no equivalent for.

7. Real-Time Streaming at Petabyte Scale

Streaming is not a native AgentCore service; real-time pipelines are provisioned and operated separately on other AWS services. Akka's streaming is built into the runtime — continuous, backpressured, **petabyte-scale, in-memory**, with no external broker — powering both agent feedback loops and high-throughput data processing (the engine behind Tubi's real-time hyper-personalization at 5 billion tokens per second).

8. For the Buyer: Maturity, Lock-In, and Accountability

AWS is a durable provider; the question is the **AgentCore product's** maturity and shape, not Amazon's viability.

Buyer concern	AWS Bedrock AgentCore	Akka
Product maturity	GA October 13, 2025; core capabilities still landing (Evaluations GA March 2026; Policy + Guardrails GA June 2026)	18 years, 100,000+ deployments, 52 banks
What you operate	Seven AgentCore services + model access + Guardrails + Knowledge Bases, integrated and run by the customer	One platform, one SLA, 24/7 SRE — Akka owns the running system
Portability / lock-in	AWS-only: IAM service-linked roles, VPC/PrivateLink, CloudFormation, CloudWatch, proprietary service APIs	Any cloud, own Kubernetes, on-prem, sovereign cloud; portable specs
Certifications & audits	Bedrock: ISO, SOC, CSA STAR L2, GDPR, FedRAMP High, HIPAA-eligible	19 standards — SOC 2 II + public SOC 3, ISO 27001/42001, EU AI Act, NIST AI RMF — plus annual pen tests, SBOMs, 40+ policies (trust.akka.io)
Delivery & outcome	Self-integrate the AgentCore services, or a systems-integration engagement; effort is guaranteed, the outcome is not	Akka Specify delivers and operates the system for one fixed price; the outcome is guaranteed
Risk transfer	Service credits on the model-API SLA	Availability and data-integrity guarantees backed by contractual indemnities
Budget predictability	Consumption meters across the AgentCore services plus model spend, plus integration labor, scaling with load	One fixed price finance can forecast

The structural point is not whether AWS will be around — it will. It is that AgentCore is a recent, consumption-metered collection of services the customer integrates and runs inside AWS, while Akka is an integrated, governed, portable platform with the SLA owned by the vendor — and, through Akka Specify, delivered and operated as a guaranteed outcome.

Customers Running Agentic and Real-Time Systems on Akka

Manulife 2,000 developers across 100 projects on one governed platform	Tubi 5B tok/s real-time hyper-personalization engine	Swiggy 71ms order-assignment AI, ~50% latency reduction	John Deere 1,000+ tractor sensors turned into real-time insight	Verizon 750% order-processing capacity gain; 6s → 2.4s response
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Common Questions

We don't have a team to integrate this — what are our options?

Two. Build it on the platform with spec-driven development, or have Akka Specify deliver and operate it for you. You provide plain-language specifications; Akka generates, governs, delivers, and runs the system — the agents, memory, streaming, APIs, and governance AgentCore leaves you to assemble — as a guaranteed outcome, for one fixed price. With AgentCore, production is self-integration or a systems-integration engagement you then own.

How is Akka Specify different from a systems-integration engagement to assemble Bedrock AgentCore into a system?

A systems-integration engagement sells effort — time-and-materials over months to wire the seven AgentCore services together — and hands back a system you own and operate; the outcome is not guaranteed. Akka Specify sells the outcome: a governed system delivered in weeks for one fixed price, then kept up, safe, and improving by Akka. You own the specifications, not the operational burden.

We're an AWS shop already. Why not just use AgentCore?

You can, but you are assembling and operating seven AgentCore services plus Bedrock model access, Guardrails, and Knowledge Bases — each billed and run separately, all locked to AWS. Akka delivers the same capabilities as one integrated runtime with a six-nines SLA Akka owns, and it runs in your AWS VPC, on another cloud, on your own Kubernetes, or on-prem. You get the platform instead of the integration project.

Sources

AgentCore GA & components: aws.amazon.com/about-aws/whats-new/2025/10/amazon-bedrock-agentcore-available — GA October 13, 2025; seven modular services (Runtime, Memory, Gateway, Identity, Code Interpreter, Browser, Observability); VPC, PrivateLink, CloudFormation at GA

AgentCore preview: aws.amazon.com/about-aws/whats-new/2025/07/amazon-bedrock-agentcore-preview — announced in preview July 16, 2025

AgentCore maturity: aws.amazon.com/about-aws/whats-new/2026/03/agentcore-evaluations-generally-available (Evaluations GA March 2026); aws.amazon.com/about-aws/whats-new/2026/06/amazon-bedrock-agentcore-policy-guardrails-generally-available (Policy + Guardrails GA June 2026)

Bedrock SLA: aws.amazon.com/bedrock/sla — 99.9% monthly uptime, measured by Bedrock API request errors per region; service credits as remedy

AgentCore governance: aws.amazon.com/bedrock/guardrails (model-layer content/PII safety); Policy = tool-access-layer enforcement

AWS Shared Responsibility Model: aws.amazon.com/compliance/shared-responsibility-model — the customer is responsible for the security, integration, and operation of everything built on top of AWS-managed infrastructure and services

AWS lock-in / VPC / IAM: docs.aws.amazon.com/bedrock-agentcore/latest/devguide/vpc-interface-endpoints.html; [.../security-iam-awsmanpol.html](https://docs.aws.amazon.com/iam/latest/reference-policy/iam-manipulating-roles.html) — IAM service-linked roles, VPC/PrivateLink, CloudWatch-

AgentCore is from AWS — isn't it the safe, mature choice?

AWS is durable; AgentCore is recent. It reached general availability in October 2025, and core governance and evaluation capabilities were still reaching GA into 2026 (Evaluations March 2026; Policy and Guardrails June 2026). Akka has 18 years and 100,000+ production deployments behind a single, integrated platform.

Can't we add EU AI Act compliance with Guardrails and Policy?

Guardrails enforce content safety at the model layer and Policy enforces tool-access rules — both useful, neither is regulation enforcement. The EU AI Act expects classification before deployment, immutable records witnessed as decisions happen, human override of running processes, and a sealed audit artifact. Akka embeds classification against 186 regulations and 877 controls, hash-chained evidence, HITL/HOTL control, and pre-deployment governance inline.

Isn't AgentCore's consumption pricing cheaper than one fixed price?

AgentCore meters its services on per-vCPU-hour, per-request, and per-token units, on top of model token spend, and the bill scales with load. Akka's shared-compute model is up to 90% cheaper to operate for the same agentic transaction volume, on one fixed price finance can forecast.

backed observability

AgentCore pricing: aws.amazon.com/bedrock/agentcore — consumption-based across the AgentCore services (per-vCPU-hour + per-GB-hour runtime, plus memory, gateway, tools, observability); Bedrock model inference billed separately; exact rates per the AWS pricing page

Bedrock compliance: aws.amazon.com/bedrock/security-privacy-responsible-ai — ISO, SOC, CSA STAR L2, GDPR, FedRAMP High, HIPAA-eligible

Akka Specify (spec-driven delivery): akka.io / akka.io/llms.txt — you provide the specifications; Akka generates, tests, governs, delivers, and operates the system as a guaranteed outcome; one fixed price covering platform, infrastructure, tokens, training, delivery, and operations; delivered in weeks; continuous improvement via reinforcement learning and distillation (up to 80% lower token cost, higher accuracy)

Akka trust center: trust.akka.io — 19 compliance standards; SOC 2 II + public SOC 3; annual pen tests, SBOMs, 40+ policies

Akka performance: akka.io/blog/go-slow-to-go-fast — Manulife up to 300% more concurrency, 30–50% faster; ~10T vs ~2T tokens/core; ~80% less compute than Python

Akka platform: 99.9999% availability, active-active HA/DR, sub-1 min RTO, zero byte RPO (contractual indemnities); 186 regulations / 877 controls / 288 with a financial penalty; 100,000+ deployments / 18 years; profitable; Dell Technologies Capital